

Amutheezan Sivagnanam

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U.S. Permanent Resident | [Scholar](#) | <https://www.linkedin.com/in/amutheezansivagnanam> | [Github](#) | Open to relocation

SUMMARY

Data Scientist with expertise in **machine learning, optimization, and operations research**. Skilled in designing, building, and deploying **end-to-end ML systems** using **Python, PyTorch, TensorFlow, and Scikit-learn**, scaling **training pipelines** with Ray and SLURM. Experienced in **translating research innovations** into **production-ready solutions**.

TECHNICAL SKILLS

- **Programming Languages:** Python, C, C++, Java, C, JavaScript
 - **Databases:** Oracle DB, MySQL, MongoDB, SQLite
 - **Data Analytics & Visualization:** NumPy, pandas, matplotlib, Spark
 - **Machine Learning & AI:** PyTorch, TensorFlow, Scikit-learn, Transformers,
 - **Software Development & Practices:** Object-Oriented, Agile Practices, Behavior Driven Development (BDD), Git
 - **Cloud & Distributed Systems:** AWS (SageMaker, S3, Lambda), Docker, Slurm, Linux
 - **Optimization:** LP, MILP, CPLEX, Gurobi, Mosek, OR-Tools
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EDUCATION

Ph.D. Computer Science, (August 2025): **Pennsylvania State University**, University Park, PA

M.S. Computer Science, (August 2022): **University of Houston**, Houston, TX

B.S. Computer Science and Engineering, (January 2018): **University of Moratuwa**, Moratuwa, Sri Lanka

EXPERIENCE

Graduate Research Assistant – Penn State University

Aug 2022 – May 2025

- Implemented an **insertion heuristic** for a dynamic routing system, **integrated** into the transit application platform to **enhance** real-time vehicle routing
- Identify the limitation in the state-of-the-art approaches with respect to implementation, and perform necessary **reorganization** (i.e., de-coupling components) and **optimization** to speed-up the computation

Graduate Research Assistant – University of Houston

Sep 2019 – Aug 2022

- Deployed an **energy-aware** scheduling algorithm into public transit operation, reducing energy costs and minimizing environmental emissions
- Implemented **automated** data-collection and cleaning framework to efficiently collect chromium vulnerability data from multiple data sources and compose into comprehensive dataset

Software Engineer – LSEG Technology

Jan 2018 – Jul 2019

- Developed and maintained application software using **Java, Python, and C++** with **Object-Oriented principles**, implementing Agile practices (Scrum) to improve team productivity, system performance, and scalability
- Designed and executed **unit testing** and **BDD-based tests** ensuring system reliability, regression compatibility, and enhanced test coverage
- Automated reporting, managed **database and back-end integrations**, and contributed to front-end development and deployment plans, streamlining operations, minimizing downtime, and improving overall system efficiency

Software Engineering Intern – WSO2 Lanka

Jul 2016 – Dec 2016

- Built outbreak alert system analyzing HL7/FHIR hospital data, supporting real-time public health monitoring
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